

ML Loop Code-Along - Step 09

Accuracy can flatter a stupid model

Incremental prerequisite prep for Imperial ML/AI coursework

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Rule: nothing appears before it is explained.

9. Accuracy can flatter a stupid model

Your spam example was exactly right: if 96 percent of emails are not spam, a classifier that always says not spam gets 96 percent accuracy while being useless at catching spam.

Step 9.1: Create a deliberately imbalanced target

Why this cell exists

This tiny demo does not train a model. It isolates the accuracy trap by creating 96 non-spam labels and 4 spam labels.

```
actual_spam = pd.Series(
    [0] * 96 + [1] * 4,
    name="actual_is_spam",
)

always_not_spam = pd.Series(
    [0] * 100,
    name="predicted_is_spam",
)
```

What to look for

Class 0 means not spam. Class 1 means spam. The prediction is intentionally stupid: always 0.

Step 9.2: Measure the stupid classifier

Why this cell exists

Accuracy rewards the model for the 96 easy majority-class cases. Recall exposes that it caught none of the spam.

```
print("Accuracy:", accuracy_score(actual_spam, always_not_spam))
print("Spam recall:", recall_score(actual_spam, always_not_spam))
```

What to look for

This is why class imbalance matters. Accuracy answered an easy question; recall answered the important one.

Checkpoint

When someone says a classifier is 96 percent accurate, ask: what are the class proportions, and what does the confusion matrix look like?